

A Review of Short Pulse Generator Technology

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Invited Paper

Abstract—Today's ultrafast, pulse generators are capable of producing high-voltage pulses, (>1 kV), with fast, leading-edge rise times, (<1 ns). A review of generator implementation methods is presented that includes a detailed discussion of the various circuit designs and a list of commercially available high-voltage pulse generators. All of these generators are capable of rise times less than a few ns and voltages greater than several hundred volts. Finally, a brief description of the three primary switch types, reed, spark gap, and solid state is presented.

Index Terms—Capacitive store, fast switching, high voltage, inductive store, Marx generators, opening switches, picosecond, pulse forming line, pulse generators, reed switches, solid state switches, spark gaps, Tesla transformer.

I. INTRODUCTION

THERE are many types of commercially available, high-voltage, ($>$ several hundred volts), nanosecond, (rise time $<$ several ns), pulse generators. These generators vary widely in performance and should be chosen according to load requirements. Examples of performance parameters are voltage output, rise time, pulsewidth, falltime, repetition rate, peak power, and average power. These generators also differ in circuit design and switch type. Circuit designs are categorized into two types of energy storage methods: 1) capacitive and 2) inductive. The most common switch types are reed, spark gap, and solid state. In most cases, the switch type determines the generator performance.

Initially, a brief discussion of pulsed power fundamentals is presented. Then, some circuit designs of selected generators are examined in detail. Finally, the three primary switch types are analyzed. Many of the circuit designs are complementary to a certain switch.

II. PULSE POWER FUNDAMENTALS

The fundamental purpose of all pulsed power systems is to convert a low-power, long-time input into a high-power, short-time output. An example of pulse compression is displayed in Fig. 1(a) and (b). The 1 kW, 1 s, input pulse is compressed into a 1 GW, 1 μ s output. In the ideal case of a lossless system, the input and output energies equal, $W_{\text{input}} = W_{\text{output}}$.

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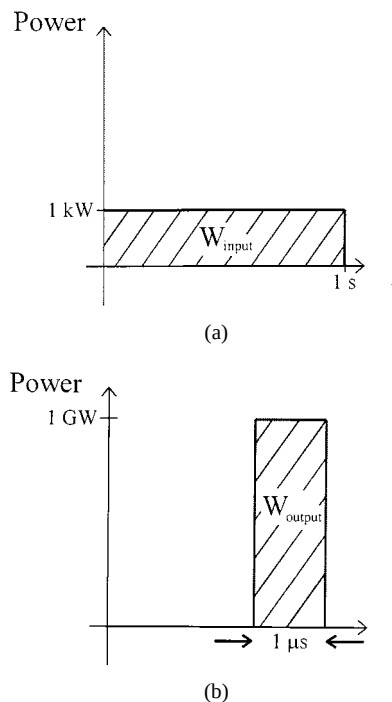


Fig. 1. Example of pulse compression: (a) input waveform and (b) output waveform.

A general pulsed power system diagram, referred to as a pulsed power train, is shown in Fig. 2. The high-voltage power supply is the power source to the system and usually plugs directly into the wall (prime power). The charging time of the energy storage section is most often determined by the specifications of the high-voltage power source and can range from less than microseconds to minutes. The energy storage section stores the pulse energy for an indeterminate amount of time, dependent upon output requirements. Pulse compression starts when the energy in the energy storage section is switched into the first pulse compression stage. The pulse compression stages perform the primary operation of compressing the pulse to an output similar to that of Fig. 1(b). The number of pulse compression stages can vary from one to many, dependent upon the application. Each of these stages usually requires a switch. The impedance match section, e.g., a transformer, is necessary to transfer maximum power to the load and to minimize reflections. The final section before the load, the primary switch, delivers the pulse to the load.

The impact that pulsed power has on biological cells, i.e., cells represented as electrical circuits, can be found in Schoenbach *et al.* [1].

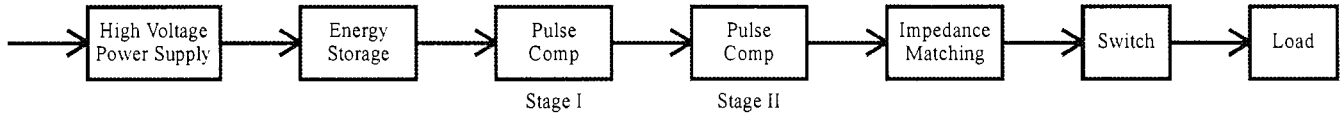


Fig. 2. Pulsed power train diagram.

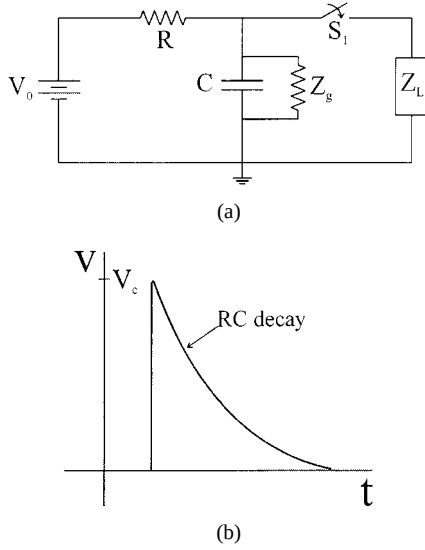


Fig. 3. (a) Simple capacitive storage discharge circuit and (b) output voltage with RC decay.

III. CIRCUIT DESIGNS

A. Capacitive Storage

There are numerous types of circuits that will provide nano- and picosecond rise time pulses. Though numerous, all of these circuits can be categorized as either capacitive or inductive store. These circuits may be implemented with any of the aforementioned switches, but some circuits are complemented by a specific switch type.

The basic capacitive energy discharge circuit and typical load voltage curve are shown in Fig. 3(a) and (b), respectively. Arrival of the pulse to the load occurs when the switch, S_1 , closes. Pulse falltime is determined by the RC time constant (for a resistive load). Ideally, the pulse rise time is zero, however, parasitic inductances from the capacitor, switch, load, and cables increase the rise time.

The pulse forming line (PFL) or transmission line section is by far the most common type circuit topology for short pulse generation, with all switch types. In fact, reed switch pulsers are almost exclusively this type of circuit. A PFL [Fig. 4(a)] of length d , is used to produce a square pulse. When the switch S_1 is closed at $t = t_0$, a pulse of amplitude $-V_0/2$ propagates down the line toward d . It then reflects at d and returns to the switch at

$$t = t_0 + \frac{2d\sqrt{\epsilon_r}}{c} \quad (\text{sec}) \quad (1)$$

where

- ϵ_r relative permittivity of the line insulator;
- d length of the transmission line in meters;
- c speed of light.

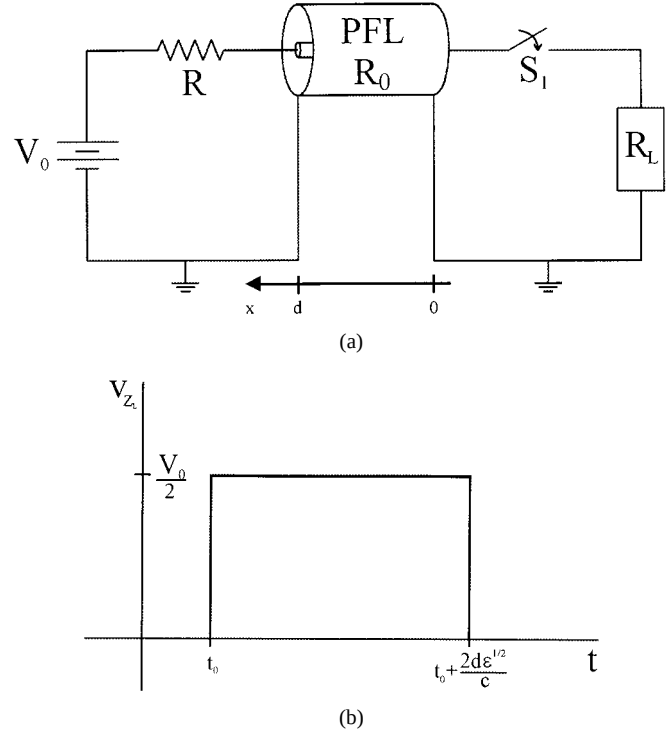


Fig. 4. (a) PFL circuit and (b) square pulse output voltage.

Upon return of the reflected pulse, the load voltage becomes zero if $R_L = R_0$, the characteristic line impedance [Fig. 4(b)]. Pulse rise and fall times are determined once again by parasitic inductances in the switch, cables, and load.

For high-voltage requirements greater than 100 kV, a Marx generator is commonly used. The "Marx" is a type of voltage multiplier invented by Marx in 1923 [2]. The basic concept of the Marx is to charge capacitors in parallel and then switch them in series, effectively increasing the voltage to $N \cdot V_c$, where N is the number of capacitors and V_c is the charging voltage. Once all the switches are closed, the Marx is said to be erected and its equivalent circuit is simply one capacitor, with a capacitance of C/N , charged to $N \cdot V_c$ (Fig. 5). Spark gaps are most often used for switching in a Marx at high power, but solid state devices such as thyristors and avalanche transistors may be used for lower power applications.

The erected Marx in Fig. 5(b) may be used as a substitute for the power supply in Fig. 4. A circuit representation of a Marx bank driven PFL is given in Fig. 6. The added inductance, L , sets the charging time of the PFL and isolates the Marx from the high-frequency load transients. If one chooses the equivalent capacitor of the PFL much lower than C_{eq} , the PFL will see a voltage doubling from the Marx. This technique was used by the authors to build a Marx driven PFL capable of producing a 700 kV pulse into a high impedance with a 400-ps rise time and a 4-ns pulsewidth [3].

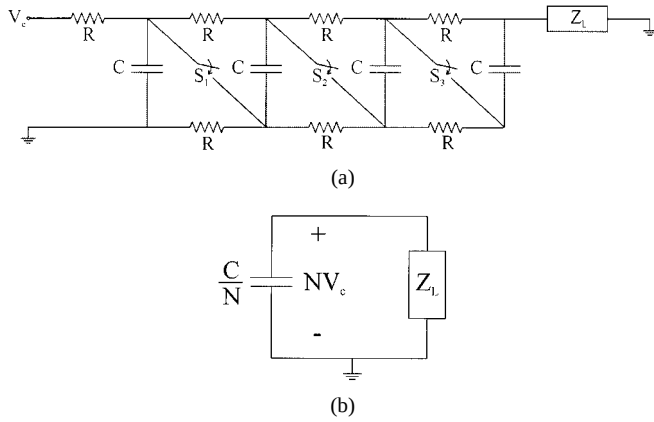


Fig. 5. (a) Marx generator circuit schematic and (b) erected Marx equivalent circuit.

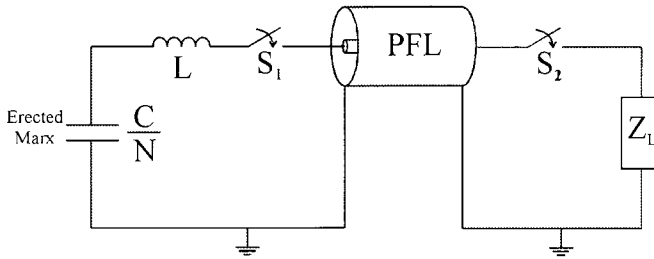


Fig. 6. Marx bank driven PFL circuit schematic.

Another device used for voltage multiplication is the Tesla transformer (Fig. 7(a)) [4]. This is a resonant transformer where $L_1 C_1 = L_2 C_2$; hence both sides of the transformer resonate at the same frequency. The output voltage V_{C2} [Fig. 7(b)] is given by

$$V_{C2} = \frac{V_0}{2} \sqrt{\frac{L_2}{L_1}} \cdot \left[\cos \left(\frac{\omega t}{\sqrt{1-k}} \right) - \cos \left(\frac{\omega t}{\sqrt{1+k}} \right) \right] \quad (\text{volts}) \quad (2)$$

where

- V_0 the charging voltage;
- ω the resonant frequency of the primary and secondary circuits;
- k the transformer coupling coefficient, and damping is neglected.

For a high coupling coefficient, $k > 0.9$, the time to peak voltage can be approximated as

$$\tau \approx \frac{\pi \sqrt{1-k}}{\omega} \quad (s). \quad (3)$$

Similar to the Marx, the Tesla transformer can be used to charge a PFL (Fig. 8). The switch, S_2 , can be fired at $0 < t < \tau$ in order to vary the applied voltage and obtain maximum voltage. This pulse generator has been implemented in a desktop version [5], capable of delivering 300 kV into a 200- Ω load, with 1 ns rise time, 4 ns pulsewidth, and rep-rates up to 100 Hz.

An interesting way to improve the rise time of the pulse from a PFL is the use of a shorting gap (Fig. 9) [6]. The output pulse for the PFL, curve A, is incident upon the shorting gap, S_2 . As the pulse passes point X, S_2 self breaks before the incident pulse

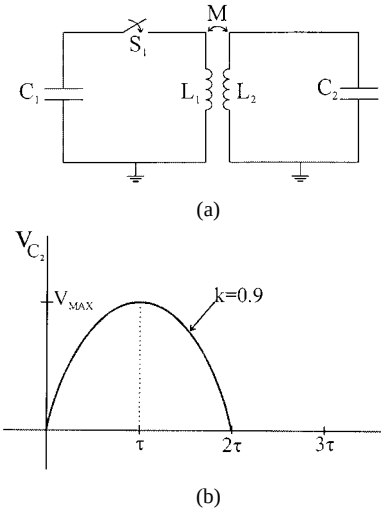


Fig. 7. (a) Tesla transformer circuit schematic and (b) output voltage with $k = 0.9$.

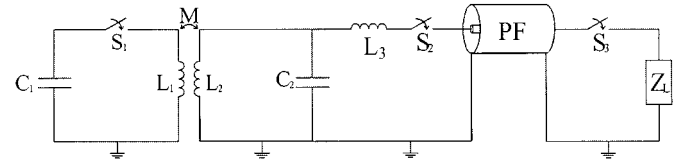


Fig. 8. Tesla transformer driven PFL.

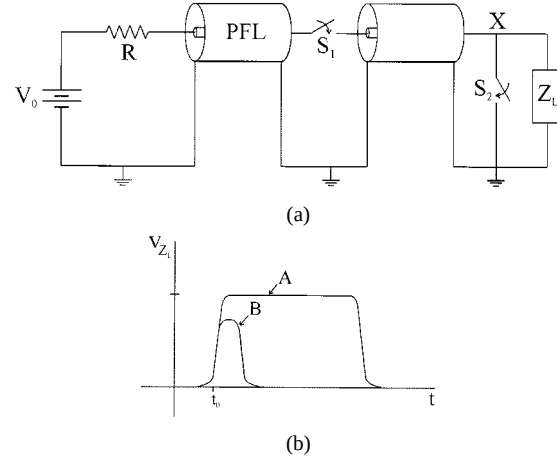


Fig. 9. (a) PFL pulse generator with shorting gap and (b) voltage curves.

reaches its peak. In effect, the incident pulse is chopped along its leading edge. The result is an output pulse with a faster rise time but at the expense of a lower amplitude voltage.

B. Inductive Storage

The other major class of nanosecond pulse generators is the inductive energy store type [7]. If one compares the energy storage density capabilities of electric and magnetic fields, one finds

$$\text{Electric Energy} \quad W_E = \frac{1}{2} \epsilon E^2 \quad (\text{J/m}^3) \quad (4)$$

and

$$\text{Magnetic Energy} \quad W_M = \frac{1}{2} \frac{B^2}{\mu} \quad (\text{J/m}^3) \quad (5)$$

where E and B are the electric and magnetic fields, respectively.

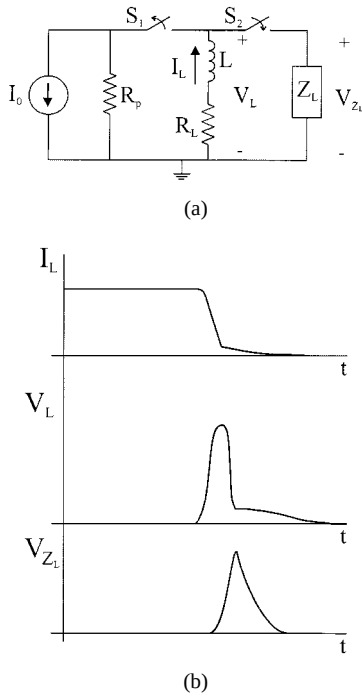


Fig. 10. (a) Simple inductive storage discharge circuit and (b) voltage and current waveforms with RL decay.

Choosing reasonable parameters of

$$\epsilon_r = 10$$

$$E = 10^8 \text{ V/m}$$

$$B = 10 \text{ T}$$

$$\mu = \mu_0$$

results in

$$W_E \approx 4.4 \times 10^5 \text{ J/m}^3$$

and

$$W_M \approx 4 \times 10^7 \text{ J/m}^3.$$

In this case the energy density of inductive storage systems is two orders of magnitude greater than that of capacitive systems.

A basic inductive energy discharge circuit schematic is shown in Fig. 10. Initially, a dc current, I_L , flows through the inductor, L . As switch S_1 opens, the current through L decreases rapidly, inducing a voltage across the inductor [Fig. 10(b)]. Near the peak of V_L , S_2 closes and the load impedance sees the output pulse.

Another example of inductive energy storage is that used by semiconductor opening switch (SOS) pulse generators [8]. The opening, or recovery, of SOS differs from traditional power diodes in that current interruption occurs in the narrow high-doped regions, not in the long drift region. Combined with a dense excess plasma in the pn junction, the SOS can switch high-density currents in nanosecond opening times. Obtainable output parameters for SOS include, voltages up to 1 MV, average powers of tens of kilowatts, peak currents of tens of kA, and a peak power of GW.

A basic circuit schematic of the SOS switching scheme is given in Fig. 11. Switches MS_1 and MS_2 are saturable core

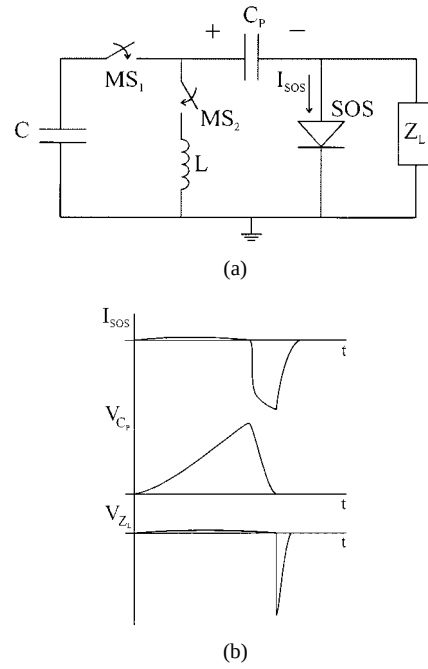


Fig. 11. (a) SOS switching scheme circuit schematic and (b) voltage and current waveforms.

inductors. Initially, MS_1 is open until the core saturates, which closes the switch. C then discharges into C_P through the SOS, which is forward biased. Eventually, the voltage across C_P peaks, at which time MS_2 closes. The current through the SOS, I_{SOS} , reverses rapidly, eventually opening the SOS. The power stored in L from the C_P discharge is then diverted to the load.

Other solid state devices used for pico and nanosecond high-power switching are drift step recovery devices (DSRD) and delayed ionization devices (DID) [9]. Unique diffusion techniques are used to achieve such fast switching times. Pulse generators which use these devices are capable of producing pulses ranging from 80 kV (into 100 Ω), 0.7 ns rise time, 2 ns pulsewidth, at 1 kHz to 20 kV (into 50 Ω), 100 ps rise time, 1.5 ns pulsewidth at 1 kHz.

One of the circuit schemes is shown in Fig. 12. Initially, both capacitors are fully charged and switches open. Then S_1 closes and C_1 discharges through I_1 and the DSRD's. The DSRD's are stacked for high-voltage holdoff. The DSRD on-resistance is very low, so I_1 will oscillate between C_1 and L_1 . At the moment I_1 crosses zero, S_2 is closed. The diodes at this time are still in a highly conducting state due to stored electron-hole pairs. The current from C_1 and C_2 will discharge in the reverse direction through the diodes, effectively doubling the current. When I_{DSRD} peaks, the initial stored charge in the diodes is depleted and the diodes open. At this time, the energy stored in L_1 and L_2 is diverted to the load, resulting in a short pulse with a rise time between 0.5 and 2 ns and a pulsewidth of L/R_L .

Another technique common to these pulsers is pulse sharpening by the silicon avalanche sharpener (SAS), Fig. 13. Normally, the SAS is biased in the blocking direction. When a pulse of high-voltage, slow rise time (a few nanoseconds) arrives at the SAS, avalanche breakdown within the drift region begins. If the incident pulse is of low enough energy (short pulsewidth), complete avalanche breakdown occurs without permanent damage

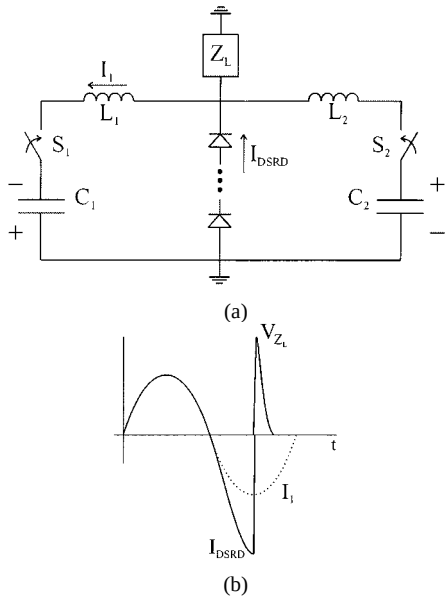


Fig. 12. (a) DSRD switching scheme circuit schematic and (b) voltage and current waveforms.

to the diode silicon structure. The output pulse has a rise time as low as 100 ps with the same voltage.

C. Pulse Generators

A list of commercially available pulse generators is given in Table I. Generators are categorized according to primary switch type. Note that all the generators produce output pulses with rise times less than several nanoseconds.

IV. SWITCH TYPES

A. Reed Switches

The reed switch, invented by Dr. W. B. Elwood at Bell Telephone Laboratories in 1936, is an interesting device [10]. It consists of two ferromagnetic, specially shaped, contact blades (reeds) positioned in a hermetically sealed glass tube (Fig. 14). The plated contact surfaces are isolated from the outside environment, which protects the contacts from contamination. The tube can be either evacuated or gas-filled. For high-voltage, fast rise time applications, it is usually back-filled with hydrogen.

The reed switch is normally open until a magnetic field applied by the external coil induces the reed tips to become temporary magnets. Once magnetic, the reed tips will move toward one another. Eventually, the E-field, a result of the voltage the switch is holding off, will become high enough to break down the gas medium between the tips, developing into an arc discharge. The arc grows in time and closes the switch.

Most commonly available high-voltage reed switches are capable of a voltage holdoff of 1.5 kV. Repetition rates, which are inversely proportional to the reed switch physical dimensions, are as high as 5 kHz at this holdoff strength. Switch lifetime varies, dependent upon the power output. Typical lifetime is 10^6 shots.

Switch rise time of industry standard reed switches is typically around a ns. However, state of the art switches exist which produce 50 ps rise time, 20 kV pulses [11]. The faster rise time

is a result of optimized gas and gas pressure parameters. One drawback of the higher voltage switch is the increase in physical size, which limits the repetition rate. For example, the 20-kV switch has a maximum repetition rate of 120 Hz.

B. Spark Gaps

High-pressure gas and oil-filled spark gaps are the switches of choice for high-power and voltage applications. Voltage holdoff can be as high as several MV and current of several MA is obtainable with several gaps operated in parallel. A brief description [12] of the switch closing process starts with the application of voltage across the gap beyond the threshold voltage (Fig. 15). This initiates an electron avalanche in the gap, which grows to the point where the space charge field is on the order of the applied field, speeding up the process via streamer growth. This growth continues until a weakly ionized channel crosses the gap. Once established, the channel begins a complex heating process until the voltage across the gap is essentially zero (compared to the holdoff voltage) and the switch is considered closed.

Spark gaps can also operate in the triggered mode by introducing a third electrode in the switch envelope. This electrode is then subjected to a short, high-voltage pulse that initiates the breakdown process. The geometrical arrangement of this third electrode has numerous variants.

The rise time of the pulse, which is the same as the falltime of the gap, is determined by two terms, the resistive and inductive phases. The inductance of the channel can be approximated as that of a thin wire of radius a ,

$$L = 2d \ln \left(\frac{b}{a} \right) \quad (\text{nH}) \quad (6)$$

where d is the gap distance in centimeters, and b is the outer conductor radius. The time constant due to this inductance is

$$\tau_L = \frac{L}{Z} \quad (\text{ns}) \quad (7)$$

where Z is the generator impedance.

The resistive phase is caused by the energy absorbed by the plasma channel as it expands. It is given as [12]

$$\tau_R = \frac{88}{Z^{1/3} E^{4/3}} \left(\frac{\rho}{\rho_0} \right)^{1/2} \quad (\text{ns}). \quad (8)$$

where E is the electric field in 10 kV/cm and ρ/ρ_0 is the ratio of gas density to that of air at STP. The total rise time of the switch is

$$\tau_{\text{tot}} = \sqrt{\tau_L^2 + \tau_R^2} \quad (\text{ns}). \quad (9)$$

Due to its low density and high pressure, hydrogen is the gas of choice for high-voltage, fast rise time, applications. Spark gaps with high-pressure hydrogen have produced MV pulses with <100 ps rise times [13].

C. Solid State Switches

The number of solid state devices which can be used as switches within pulse generators is numerous and includes thyristors, bipolar junction transistors

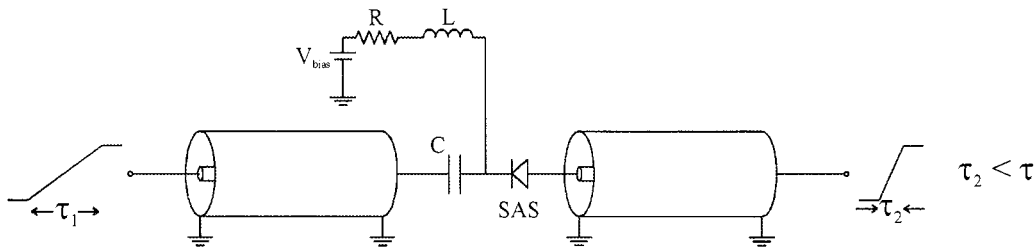


Fig. 13. SAS circuit schematic.

TABLE I
EXAMPLES OF COMMERCIALY AVAILABLE SHORT PULSE GENERATORS

Company	Model	Primary Switch	Output Voltage	Load (ohm)	Risetime	Pulsewidth	Rep-Rate
Barth Electronics	632	Reed	+/- 500 V to +/- 2.5 kV	50	50 ps	750 ps to 5 ns	single shot or 0.1 to 10 Hz
Barth Electronics	732	Reed	< 100 V to 2 kV	50	< 50 ps	750 ps to 1 μ s	.05 to 120 Hz
Kentech	CPSx	FET and SAS	up to 6 kV	50	< 150 ps	2 ns	10 Hz to 1 kHz
Kentech	PGBx	FET and SAS	up to 12 kV	50	< 90 ps	3 ns	100 Hz
Moose-Hill PSG	NPGx	DSRD	up to 30 kV	50	.6 to 4 ns	1-20 ns	up to 100 kHz
Moose-Hill PSG	PPGx	DSRD and SAS	2-20 kV	50	50-100 ps	.2-100 ns	up to 25 kHz
Integrated Technologies	SEF	Spark Gap	20-200 kV	50	1 ns	4 ns	single shot to 100 Hz
Integrated Technologies	SEF w/slicer	Spark Gap	20-120 kV	50	100 ps	500 ps	single shot to 100 Hz

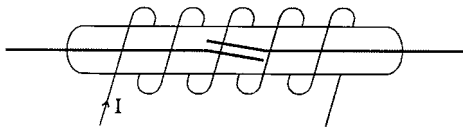


Fig. 14. Typical reed switch configuration.

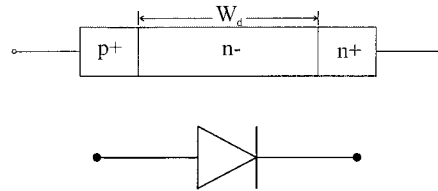


Fig. 16. Power diode structure.

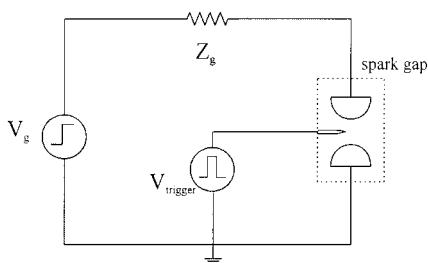


Fig. 15. Circuit schematic of a spark gap switch.

(BJT's), metal-oxide-semiconductor field effect transistors (MOSFET's), insulated gate bipolar transistors (IGBT's), and numerous diode types. It is not the authors' intention to explain the details of operation of all these devices. However, an understanding of the pn junction, the building block of all power semiconductors, will make all other power devices more clearly understood.

A practical pn junction diode is shown in Fig. 16 [14]. It consists of a heavily doped n+ type substrate upon which a lightly doped n-epitaxial layer is grown. The p+ region is formed by heavy p type diffusion into the n-region. The n-layer, often

called the drift region, W_d , is the prime feature not found in low-power diodes. Its function is to absorb the depletion layer of the reverse biased diode. It is the drift region which establishes the reverse breakdown voltage.

A power diode requires a certain amount of time to switch from reverse to forward bias and vice versa. Turn-on starts with the diode reverse biased, at which time the depletion layer, located primarily in the drift region, contains stored space charge. In order to forward bias the diode, this space charge must be removed, which can be modeled as the discharge of a capacitor. In fact, the depletion layer during reverse bias is often modeled as a parallel plate capacitor. As a result, the turn-on time for these devices is specified for a current time rate of change, di_f/dt .

Turnoff time is essentially the reverse process of turn-on. During the on state, the drift region has stored excess carriers. These excess carriers must be removed before the diode metallurgical junctions can be reversed biased. The forward current will decrease at a rate of di_r/dt . After the current goes negative, excess carriers are swept out of the drift region. Eventually, the junction becomes reverse biased and the diode voltage rapidly becomes negative, at a much faster rate than di_r/dt . This rapid

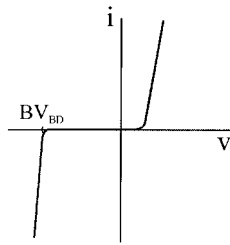


Fig. 17. Power diode V - I characteristic.

recovery makes the power diode better suited as an opening switch for ps pulse applications.

As the reverse bias across a pn junction is increased, the current will rapidly increase at the breakdown voltage, BV_{BD} (Fig. 17). Current growth at BV_{BD} is due to impact ionization in the depletion layer. One free electron gains enough energy from the applied E-field to strike a silicon atom hard enough to free another electron from its covalent bond. This is termed impact ionization. This process of freeing electrons can increase rapidly in a high enough E-field and cascade to an avalanche of free electrons, essentially turning the diode on in the reverse direction. If left to continue for a significant time, power dissipation will destroy the device. However, for very short pulses, this process can be used in sharpening applied pulses with no damage done to the lattice.

Some of the devices used for subnanosecond switching are the semiconductor opening switch (SOS), drift step recovery device (DSRD), GaAs thyristor, and inverse recovery diode (IRD). All these devices, except for the GaAs thyristor are specially constructed power diodes capable of producing pulses with rise times between 100 ps and 1 ns. Voltage capabilities of these individual devices range from one to tens of kilovolts, however, devices may be stacked to obtain higher voltages.

D. Summary

A brief overview of high-voltage nanosecond pulser generator technology has been presented. Circuit design types were divided into capacitive and inductive storage types and examined in some detail. Examples of commercially available nanosecond and picosecond pulsers are given. The characteristics of three primary switch types (reed, spark gap, and solid state) were described, including turn-on time, holdoff voltage, and turnoff.

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